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ARRAY SUBSTRATE

Abstract

An array substrate includes a substrate, a first pixel electrode, a second pixel electrode, a first scan line, a second scan line and a data line. The first and second pixel electrodes are disposed on the substrate and arranged along a first direction. The first and second scan lines are disposed on the substrate and extend along the first direction, wherein the first and second pixel electrodes are disposed between the first and second scan lines. The first and second traces are disposed on the substrate and extend along a second direction different from the first direction, wherein the first and second traces are electrically connected to the first and second scan lines, respectively, the first trace crosses and is electrically insulated from the second scan line, and the data line is disposed between the first trace and the second trace.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to an array substrate, and more particularly to an array substrate applied to an electronic device with narrow border.

2. Description of the Prior Art

[0002] In order to meet requirements on appearance designs and space utilization, display devices have been developed to have reduced border width. At present, it has been developed to use traces formed of the same conductive layer with the data lines to bridging scan lines, and the traces are disposed in the display region. However, this approach will increase the number of signal lines in the display region, thus increasing the coupling capacitance between adjacent signal lines and further adversely affecting the display quality. Therefore, how to meet the requirement on reduction of border width while maintaining the display quality is one of the development objectives for those skilled in the art.

SUMMARY OF THE INVENTION

[0003] An embodiment of the present invention provides an array substrate, which includes a substrate, a first pixel electrode, a second pixel electrode, a first scan line, a second scan line and a data line. The first pixel electrode and the second pixel electrode are disposed on the substrate and arranged along a first direction. The first scan line and the second scan line are disposed on the substrate and extend along the first direction, wherein the first pixel electrode and the second pixel electrode are disposed between the first scan line and the second scan line. The first trace and the second trace are disposed on the substrate and extend along a second direction different from the first direction, wherein the first trace is electrically connected to the first scan line, the second trace is electrically connected to the second scan line, the first trace crosses and is electrically insulated from the second scan line, and the first trace and the second trace are formed of a first conductive layer. The data line is disposed on the substrate and extends along the second direction, wherein in the top view of the array substrate the data line is disposed between the first pixel electrode and the second pixel electrode, the data line crosses the second scan line, the data line is formed of a second conductive layer different from the first conductive layer, and the data line is disposed between the first trace and the second trace.

[0004] Another embodiment of the present invention provides an array substrate, which includes a substrate, a first scan line, a second scan line, a first trace, a second trace and a data line. The first scan line and the second scan line are disposed on the substrate and extend along the first direction, wherein each of the first scan line and the second scan line comprises a plurality of line sections and a plurality of first bridging sections. The first trace and the second trace are disposed on the substrate and extend along a second direction different from the first direction, wherein the first trace is electrically connected to the first scan line, and the second trace is electrically connected to the second scan line. The data line is disposed on the substrate and extends along the second direction, wherein the data line crosses one of the plurality of first bridging sections of the first scan line and one of the plurality of first bridging sections of the second scan line. A number of the plurality of first bridging sections of the first scan line is the same as a number of the plurality of first bridging sections of the second scan line. The one of the first bridging sections of the second scan line in the data line in the top view of the array substrate overlaps the data line and the first trace.

[0005] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 schematically illustrates a top view of an array substrate according to a first embodiment of the present invention.

[0007] FIG. 2 schematically illustrates a cross-sectional view of FIG. 1 taken along a line A-A'.

[0008] FIG. 3 schematically illustrates a top view of an array substrate according to a second embodiment of the present invention.

[0009] FIG. 4 schematically illustrates a cross-sectional view of FIG. 3 taken along a line B-B'.

[0010] FIG. 5 schematically illustrates a top view of an array substrate according to a variation of the second embodiment of the present invention.

[0011] FIG. 6 schematically illustrates a top view of an array substrate according to another variation of the second embodiment of the present invention.

[0012] FIG. 7 schematically illustrates a top view of an array substrate according to a third embodiment of the present invention.

[0013] FIG. 8 schematically illustrates a cross-sectional view of FIG. 7 taken along a line C-C'.

[0014] FIG. 9 schematically illustrates a top view of an array substrate according to a variation of the third embodiment of the present invention.

[0015] FIG. 10 schematically illustrates a top view of an array substrate according to another variation of the third embodiment of the present invention.

DETAILED DESCRIPTION

[0016] The contents of the present invention will be described in detail with reference to specific embodiments and drawings. It is noted that, for purposes of illustrative clarity and being easily understood by the readers, the following drawings may be simplified schematic diagrams, and elements therein may not be drawn to scale. The numbers and dimensions of the elements in the drawings are just illustrative and are not intended to limit the scope of the present invention.

[0017] In the present invention, the array substrate may be applied to an electronic device that needs a pixel array. For example, the electronic device may be a display device, an image sensing device, an X-ray sensing device, or any other suitable device. The display device may be, for example, a non-self-luminous display device or a self-luminous display device. The non-self-luminous display device may include, for example, a liquid crystal display device or any other suitable device. The self-luminous display device may include, for example, an organic light-emitting diode display device, an inorganic light-emitting diode display device, or any other suitable device. Depending on the types of display devices, the display device may further include another substrate and/or elements in addition to the array substrate. For example, if the display device is the liquid crystal display device, the display device may include the array substrate, a color filter substrate and a liquid crystal layer, and the liquid crystal layer may be disposed between the color filter substrate and the array substrate, but it is not limited thereto. If the display device is the self-luminous display device, the display device may include the array substrate, light emitting diodes and an encapsulation layer, in which the light emitting diodes and the encapsulation layer are disposed on the array substrate, but it is not limited thereto. The array substrate in the following description is applied to a display device as an example, but it is not limited thereto.

[0018] Refer to FIG. 1 and FIG. 2, in which FIG. 1 schematically illustrates a top view of an array substrate according to a first embodiment of the present invention, and FIG. 2 schematically illustrates a cross-sectional view of FIG. 1 taken along a line A-A'. For clarity, FIG. 1 and FIG. 2 only show some of elements included in the array substrate, but the array substrate of the present invention is not limited to the configuration as shown. As shown in FIG. 1, the array substrate 1 is applied to the display device and may include an active region AA and a peripheral region PR, wherein the peripheral region PR is disposed at at least one side of the active region AA. In an embodiment as illustrated in FIG. 1, the peripheral region PR may surround the active region AA, but not limited thereto. The active region AA of the array substrate 1 may, for example, correspond

to a display region of the display device, but it is not limited thereto.

[0019] In an embodiment as illustrated in FIG. 1, the array substrate 1 may be a dual-gate type of array substrate, but it is not limited thereto. In detail, the array substrate 1 may include a substrate 12, a plurality of pixel electrodes 14, a plurality of scan lines 16, a plurality of traces 18, a plurality of data lines 20 and a plurality of switch elements 22, wherein the pixel electrodes 14, the scan lines 16, the traces 18, the data lines 20 and the switch elements 22 are disposed on the substrate 12 in the active region AA. The substrate 12 may be, for example, a flexible substrate, or a non-flexible substrate. The substrate 12 may include, for example, glass, quartz, sapphire, acrylic, polyimide (PI), any other suitable material, or a combination of the above-mentioned materials, but it is not limited thereto.

[0020] As shown in FIG. 1, each of the pixel electrodes 14 may be electrically connected to a corresponding one of the switch elements 22. If the array substrate 1 is applied to a display device, pixel voltages to be applied to the pixel electrodes 14 may be changed through the switch elements 22, thereby adjusting brightness of pixels or sub-pixels of the display device, but it is not limited thereto. In an embodiment as illustrated in FIG. 1, the pixel electrodes 14 may be arranged in the active region AA as an array, wherein each row of the array may extend along a first direction D1, and each column of the array may extend along a second direction D2, which is different from the first direction D1, e.g., perpendicular to the first direction D1. However, it is not limited thereto. For example, the pixel electrodes 14 may include a pixel electrode 14a, a pixel electrode 14b, a pixel electrode 14c and a pixel electrode 14d, wherein the pixel electrode 14a and the pixel electrode 14b are arranged along the first direction D1, and the pixel electrode 14c and the pixel electrode 14d are arranged along the first direction D1, the pixel electrode 14a and pixel electrode 14c are arranged along the second direction D2, and the pixel electrode 14b and pixel electrode 14d are arranged along the second direction D2. The arrangement of the pixel electrodes 14 according to the present invention is not limited to the above-mentioned one.

[0021] The scan lines 16 respectively extend along the first direction D1 and are arranged along the second direction D2. In this embodiment, the scan lines 16 are grouped into a plurality of scan line pairs 16P, which are arranged along the second direction D2. Each of the scan line pairs 16P consists of two adjacent scan lines 16. A row of pixel electrodes 14 is disposed between two of the scan lines 16, which belong to the same scan line pair 16P, so that the pixel voltages applied to the pixel electrodes 14 in the same row may be changed through the scan lines 16 of the same scan line pair 16P, but it is not limited thereto. In this embodiment, the pixel voltages applied to the pixel electrodes 14 in the same row and connected to the same data line 20 may be changed sequentially through different scan lines 16 of the same scan line pair 16P. For example, the scan lines 16 may include a scan line 16a, a scan line 16b, a scan line 16c and a scan line 16d, wherein the scan line 16a and the scan line 16b constitute one scan line pair 16P, and the scan line 16c and the scan line 16d constitute another scan line pair 16P. The pixel electrode 14a and the pixel electrode 14b are disposed between the scan line 16a and the scan line 16b, and the pixel voltages of which are changed through the scan line 16a and the scan line 16b, respectively. The pixel electrode 14c and the pixel electrode 14d are disposed between the scan line 16c and the scan line 16d, and the pixel voltages of which are changed through the scan line 16c and the scan line 16d, respectively. In other words, counting the pixel electrodes 14 in the same row from left to right of the active region AA, the odd-numbered pixel electrodes 14 may be controlled through the scan line 16a, and the even-numbered pixel electrodes 14 may be controlled through the scan line 16b, but it is not limited thereto.

[0022] The traces 18 are electrically connected to the scan lines 16, respectively and extend from the corresponding scan lines 16 to the peripheral region PR at a side, e.g., a first side S1, of the active region AA along the second direction, thereby electrically connecting the scan lines 16 to an element, e.g., a gate driving circuit, in the peripheral region PR. In this case, the traces 18 may cross and be electrically insulated from the scan lines 16 disposed between the corresponding scan

lines **16** and the first side **S1** of the active region **AA**. For example, the traces **18** may include a trace **18a** and a trace **18b**, which are electrically connected to the scan line **16a** and the scan line **16b**, respectively, and the trace **18a** crosses and is electrically insulated from the scan line **16b**. It is to be noted that since the traces **18** electrically connected to the corresponding scan lines **16** may extend to the peripheral region **PR** along the second direction **D2**, it is not necessary to form traces at two ends of one of the scan lines **16**. Accordingly, the number of the traces at left side and right side of the scan lines **16** may be reduced, so that the width of the peripheral region **PR** at a second side **S2** of the active region **AA** and the width of the peripheral region **PR** at a third side **S3** of the active region **AA** may be reduced. Therefore, the display device may have a narrow border. In some embodiments, one of the traces **18** may cross the pixel electrodes **14** in the same column and disposed between the corresponding scan line **16** and the first side **S1** of the active region **AA**. [0023] In the embodiment as illustrated in FIG. 1, the lengths of the traces **18** in the second direction **D2** may differ from each other as distances from the scan lines **16** to the first side **S1** of the active region **AA**. In an embodiment, the traces **18** are gradually shorter along a specified direction, e.g., the first direction **D1**. For example, longer traces **18** are closer to the second side **S2** or the third side **S3** of the active region **AA**, but it is not limited thereto. In some embodiments, the sequence of arranging the traces **18** in the first direction **D1** may not depend on the sequence of the sizes in length of the traces.

[0024] In the embodiment as illustrated in FIG. 1, at least one of the scan lines **16** may include at least two line sections **161** and at least one bridging section **162** electrically connected between the line sections **161**. In other words, the line sections **161** and the bridging section(s) **162** are interconnected to form the scan line **16**. For example, each of the scan lines **16** disposed between the first side **S1** of the active region **AA** and the scan line **16** (e.g., the scan line **16a** shown in FIG. 1) farthest from the first side **S1** may include at least one bridging section **162**. In an embodiment, the scan lines **16** disposed between two adjacent rows of pixel electrodes **14** may include the same number of line sections **161** and the same number of bridging section **162**, but it is not limited thereto. For example, each of the scan line **16b** and the scan line **16c** may include two line sections **161** and one bridging section **162**, each of the scan line **16d** and the scan line **16e** may include three line sections **161** and two bridging sections **162**, and so on.

[0025] As shown in FIG. 2, the traces **18** and the bridging sections **162** may be formed of different conductive layers. Therefore, the traces **18** may overlap the bridging sections **162** in the top view of the array substrate **1**, such that the traces **18** can cross the scan lines **16** while being electrically insulated from the scan lines **16**. For example, the line sections **161** and the traces **18** are formed of a conductive layer **C1**, and the bridging sections **162** are formed of a conductive layer **C3** different from the conductive layer **C1**, so that the bridging section **162** of the scan line **16b** may cross and overlap the trace **18a**. Under this circumstance, the bridging section **162** of the scan line **16c** as shown in FIG. 1 may cross the trace **18a** and the trace **18b**, and so on. In an embodiment, each of the line sections **161** and the bridging section(s) **162** may be of a strip shape, but it is not limited thereto. In this disclosure, the top view of the array substrate may, for example, be viewed along a top view direction **TD** perpendicular to a top surface **12S** of the substrate **12**, as shown in FIG. 2.

[0026] In some embodiments, the scan lines **16** may respectively include the same number of line sections **161** and the same number of bridging sections **162**. In an example, the number of bridging sections **162** included in each scan line **16**, as shown in FIG. 3, may be the same as a number of the data lines **20**. By designing the scan lines **16** with the same number of bridging sections **162**, the resistances of the scan lines **16** may be close to each other, so as to avoid abnormality of the display device and improve displaying quality. Under this circumstance, in some embodiments, the traces **18** in the active region **AA** may have the same length, and each trace **18** may cross the scan lines **16**, which are not electrically connected thereto, thereby unifying the coupling capacitances of the scan lines **16**. However, it is not limited thereto.

[0027] As shown in FIG. 1, the data lines **20** extend along the second direction **D2** and cross the

scan lines **16**, and the data lines **20** are electrically insulated from the scan lines **16**. In the embodiments as illustrated in FIG. **1** and FIG. **2**, the data lines **20** may be formed of a conductive layer **C2** different from the conductive layer **C1** and the conductive layer **C3**, but it is not limited thereto. The bridging section **162** or the line sections **161** of one of the scan lines **16** may cross one of the data lines **20** while being electrically insulated from the data line **20**. For example, the data lines **20** may include a data line **20a** and a data line **20b**, which are arranged along the first direction **D1**, and the bridging section **162** of the scan line **16b** or **16c** may cross the data line **20a**. Furthermore, one of the bridging sections **162** of the scan line **16d** or **16e** may cross the data line **20a**, and the other of the bridging sections **162** of the scan line **16d** or **16e** may cross the data line **20b**. In some embodiments, the scan lines **16** respectively include the same number of line sections **161** and the same number of bridging sections **162**. Meanwhile, each of the data lines **20** may cross one of the bridging sections **162** of each scan line **16**, but it is not limited thereto.

[0028] In an embodiment, one of the data lines **20** may be disposed between two adjacent columns of the pixel electrodes **14** in the top view of the array substrate **1** and electrically connected to the pixel electrodes **14** in the two adjacent columns of the pixel electrodes **14**. Therefore, each data line **20** may be used for providing the pixel voltage of two adjacent columns of the pixel electrodes **14**, thereby reducing the number of data lines **20**. On the other hand, two columns of pixel electrodes **14** may be disposed between two adjacent data lines **20**. For example, the data line **20a** may be disposed between the pixel electrode **14a** and the pixel electrode **14b** and between the pixel electrode **14c** and the pixel electrode **14d** and be electrically connected to the pixel electrodes **14a-14d**. In addition, the pixel electrode **14b** and the pixel electrode **14e** arranged along the first direction **D1** are disposed between two adjacent data lines **20a** and **20b**. Furthermore, in the top view of the array substrate **1**, one of the data lines **20** may be disposed between two of the traces **18** electrically connected to the scan lines **16** belonging to the same scan line pair **16P**. For example, the data line **20a** may be disposed between the trace **18a** and the trace **18b**, but it is not limited thereto.

[0029] As shown in FIG. **1**, the gate, source and drain electrodes of each of the switch elements **22** are electrically connected to a corresponding one of scan lines **16**, a corresponding one of data lines **20** and a corresponding one of pixel electrodes **14**, respectively. For example, the switch elements **22** may include a switch element **22a**, a switch element **22b**, a switch element **22c** and a switch element **22d**. The gate, source and drain electrodes of the switch element **22a** are electrically connected to the scan line **16a**, the data line **20a** and the pixel electrode **14a**, respectively; the gate, source and drain electrodes of the switch element **22b** are electrically connected to the scan line **16b**, the data line **20a** and the pixel electrode **14b**, respectively; the gate, source and drain electrodes of the switch element **22c** are electrically connected to the scan line **16c**, the data line **20a** and the pixel electrode **14c**, respectively; and the gate, source and drain electrodes of the switch element **22d** are electrically connected to the scan line **16d**, the data line **20a** and the pixel electrode **14d**, respectively.

[0030] As shown in FIG. **1** and FIG. **2**, in the top view of the array substrate **1**, in a case that one of the traces **18** disposed between one of the switch elements **22** and a corresponding one of the data lines **20**, the source electrode of the switch element **22** may overlap the trace **18**. For example, the source electrode of the switch element **22a** may cross and overlap the trace **18a**. The source electrode of the switch element **22c** may cross and overlap the trace **18a**. The source electrode of the switch element **22d** may cross and overlap the trace **18b**.

[0031] In the embodiment as illustrated in FIG. **1** and FIG. **2**, the gate electrodes of the switch elements **22** may be formed of the same conductive layer **C1** as the line sections **161** of the scan lines **16**. The source electrodes and drain electrodes of the switch elements **22** may be formed of the same conductive layer **C2** as the data lines **20**. However, it is not limited thereto. Each of the switch elements **22** may further include a semiconductor layer and a gate insulating layer (not shown), wherein the gate insulating layer is disposed between the gate electrode and the

semiconductor layer and opposite sides of the semiconductor layer are in contact with the source electrode and the drain electrode, respectively. The switch elements **22** are, for example, bottom gate thin film transistors, but it is not limited thereto. In some embodiments, the switch elements **22** may be other types of thin film transistors, depending on design requirements.

[0032] As shown in FIG. **2**, the conductive layer **C1**, the conductive layer **C2** and the conductive layer **C3** are sequentially disposed on the substrate **12**. An insulating layer **IN1** is disposed between the conductive layer **C1** and the conductive layer **C2**, and an insulating layer **IN2** is disposed between the conductive layer **C2** and the conductive layer **C3**. The insulating layer **IN1** and the insulating layer **IN2** may have a through hole **TH**, through which the bridging section **162** may be electrically connected to the corresponding line section **161**. Since the traces **18** are formed of the conductive layer **C1**, and the data lines **20** are formed of the conductive layer **C2**, the insulating layer **IN1** may also exist between the traces **18** and the data lines **20** to reduce the coupling capacitances between the traces **18** and the data lines **20**, thereby minimizing abnormality of the display device. In an embodiment, the resistivity of the conductive layer **C1** and the resistivity of the conductive layer **C2** are lower than the resistivity of the conductive layer **C3**. For example, the conductive layer **C1** and the conductive layer **C2** may include metal, and the conductive layer **C3** may include transparent conductive material, but it is not limited thereto. The transparent conductive material may, for example, include indium tin oxide (ITO) or any other suitable material. In some embodiments, the pixel electrodes **14** may be formed of the conductive layer **C3**, but it is not limited thereto.

[0033] In some embodiments, the array substrate **1** may further include common lines, e.g., common lines **24** as shown in FIG. **3**, which are disposed on the substrate **12** in the active region **AA**, for transmitting a common voltage. The common lines may, for example, extend along the second direction **D2** and formed of the conductive layer **C1**, but it is not limited thereto. In the top view of the array substrate **1**, one of the common lines may overlap the pixel electrodes **14** in the same column, but it not limited thereto.

[0034] It is understood from the above description that in this embodiment of dual-gate type of array substrate **1**, since the number of the traces **18** disposed in the peripheral region **PR** at the second side **S2** and the third side **S3** of the active region **AA** can be significantly reduced, and the coupling capacitances between the traces **18** and the data lines **20** can also be reduced, the array substrate **1** can meet the requirement on narrow border while improving displaying quality with a reduced number of data lines **20**.

[0035] The array substrate is not limited to the above embodiments and may include other embodiments. To simplify the description, other embodiments will be labeled with the same reference numerals as the first embodiment. In order to easily compare the differences between the first embodiment and other embodiments, the differences between different embodiments will be mentioned below, and repeated parts will not be described redundantly.

[0036] FIG. **3** schematically illustrates a top view of an array substrate according to a second embodiment of the present invention. FIG. **4** schematically illustrates a cross-sectional view of FIG. **3** taken along a line B-B'. As shown in FIG. **3** and FIG. **4**, the array substrate **2a** in this embodiment differs from the array substrate **1** as illustrated in FIG. **1** in the coupling manner of the pixel electrodes to the scan lines **16** and the data lines **20** through the switch elements **22**. In detail, the array substrate **2a** may include a substrate **12**, a plurality of pixel electrodes **14**, a plurality of scan lines **16**, a plurality of traces **18**, a plurality of data lines **20** and a plurality of switch elements **22**, wherein the pixel electrodes **14**, the scan lines **16**, the traces **18**, the data lines **20** and the switch elements **22** are disposed on the substrate **12** in the active region **AA**. Each of the pixel electrodes **14** is electrically connected to a drain electrode of a corresponding one of the switch elements **22**, but it is not limited thereto. The pixel electrodes **14** in this embodiment are arranged in a manner similar to the arrangement of the pixel electrodes **14** shown in FIG. **1**, and thus it will not be described redundantly.

[0037] In the embodiment as illustrated in FIG. 3, the scan lines **16** are arranged along the second direction D2, and the pixel electrodes **14** in each row are alternately arranged with the scan lines **16** along the second direction D2. In this case, one of the pixel electrodes **14** may be disposed in a region surrounded by two adjacent scan lines **16** and two adjacent data lines **20**. Furthermore, each scan line **16** may be electrically connected to the gate electrodes of the switch elements **22**, which correspond to the pixel electrodes **14** in the same row, and each data line **20** may be electrically connected to the source electrodes of the switch elements **22**, which correspond to the pixel electrodes **14** in the same column, so that the pixel electrodes **14** may be electrically connected to the corresponding data lines **20** through the switch elements **22**. For example, the pixel electrode **14a** and the pixel electrode **14b** may be arranged along the second direction D2, the pixel electrode **14a** may be electrically connected to the data line **20a** through the switch element **22a**, and the pixel electrode **14b** may be electrically connected to the data line **20a** through the switch element **22b**. However, it is not limited thereto.

[0038] As shown in FIG. 3, in the top view of the array substrate **2a**, the scan lines **16** extend along the first direction D1, and the traces **18** extend along the second direction D2, wherein each of the traces **18** is electrically connected to a corresponding one of the scan lines **16**, and at least one of the traces **18** crosses and is electrically insulated from at least one non-corresponding scan line **16**, i.e., at least one of the scan lines **16** that is not electrically connected to the at least one of the traces **18**. For example, the trace **18a** crosses the scan line **16b** and is electrically insulated from the scan line **16b**. In the embodiment as illustrated in FIG. 3, each scan line **16** may include a plurality of line sections **161** and a plurality of bridging sections **162**, and the scan lines **16** may have the same number of bridging sections **162**, so as to unify the resistances of the scan lines **16**. In an embodiment, the bridging sections **162** of the scan lines **16** may have the same length in the first direction D1, but it is not limited thereto.

[0039] In this embodiment, the traces **18** and the data lines **20** are alternately arranged along the first direction D1, but it is not limited thereto. For example, one trace **18** and one data line **20** are disposed between two adjacent columns of pixel electrodes **14**, but it is not limited thereto. As shown in FIG. 3, the data lines **20** may be disposed at right sides of corresponding columns of pixel electrodes **14**, respectively, but it is not limited thereto. In some embodiments, at least one trace **18** may be disposed between two adjacent data lines **20**, but it is not limited thereto.

[0040] In the embodiment as illustrated in FIG. 3, the traces **18** may have different lengths along the second direction D2 in the top view of the array substrate **2a**, so that at least one of the scan lines **16** may not be overlapped with at least one of the traces **18**, which are not electrically connected thereto, but it is not limited thereto. Under this circumstance, at least a portion of the bridging sections **162** of the scan lines **16** may overlap the traces **18** and the data lines **20**, which are not electrically connected thereto. For example, the scan line **16a** may not be overlapped with the trace **18b**, and the scan line **16b** may overlap the trace **18a** and the data line **20a**, which are not electrically connected thereto. In some embodiments, the bridging sections **162** of the scan lines **16** may not be overlapped with the traces **18**, and under this circumstance, the scan lines **16** may further include bridging sections **163**, as shown in FIG. 7.

[0041] Similar to the above-described embodiments, the traces **18** in the embodiment as illustrated in FIG. 3 may be arranged along the first direction D1 according to the sequence of the sizes in the lengths of the traces **18**. For example, longer traces **18** are closer to the second side S2 of the active region AA, but it is not limited thereto. In some embodiments, the sequence of arranging the traces **18** in the first direction D1 does not follow a length rule.

[0042] As shown in FIG. 3 and FIG. 4, the line sections **161** of the scan lines **16** may be formed of the conductive layer C1, the data lines **20** may be formed of the conductive layer C2, and the bridging sections **162** may be formed of the conductive layer C3. Therefore, the bridging sections **162** may be electrically connected to adjacent line sections **161** while being electrically insulated from the traces **18** and the data lines **20**. As shown in FIG. 4, an insulating layer IN1 may be

disposed between the conductive layer C1 and the conductive layer C2, and an insulating layer IN2 may be disposed between the conductive layer C2 and the conductive layer C3. In this embodiment, the insulating layer IN1 may be used as, for example, the gate insulating layer, but not limited thereto. The sequence to stack conductive layers C1-C3, the insulating layer IN1 and the insulating layer IN2 may be similar to or the same as that illustrated in FIG. 2. Likewise, the coupling capacitances between the traces 18 and the data lines 20 may be reduced accordingly, so as to minimize abnormality of the display device in this embodiment, and thus it will not be described redundantly. Furthermore, the materials of the conductive layer C1, the conductive layer C2 and the conductive layer C3 may be the same as those of the conductive layer C1, the conductive layer C2 and the conductive layer C3 described in the first embodiment, respectively, and thus it will not be described redundantly.

[0043] It is to be noted that since the bridging sections 162 are formed of the conductive layer C3, and the resistivity of the conductive layer C3 is greater than resistivity of the conductive layer C1 for forming the line sections 161, it is advantageous to unify resistances of the scan lines 16 by making the scan lines 16 have the same number of bridging sections 162, thereby minimizing signal variation with locations and improving displaying quality of images. Furthermore, since the traces 18 and the bridging sections 162 are formed of different conductive layers C1 and C3, respectively, interference among signals on different scan lines may be mitigated.

[0044] As shown in FIG. 3 and FIG. 4, the array substrate 2a may further include a plurality of common lines 24 disposed on the substrate 12 in the active region AA and for transmitting a common voltage. The common lines 24 may, for example, extend along the second direction D2 and formed of the conductive layer C2, but it is not limited thereto. The common lines 24 in the top view of the array substrate 2a may overlap the pixel electrodes 14 in the same column, so as to form storage capacitors with the pixel electrodes 14, but it not limited thereto. In some embodiments, the common lines 24 may be formed of a conductive layer other than the conductive layers C1-C3. Other portions of the array substrate 2a shown in FIG. 3 and FIG. 4 may be similar to or the same as those included in the array substrate described above or below, and thus it will not be described redundantly.

[0045] FIG. 5 schematically illustrates a top plan view of an array substrate according to a variation of the second embodiment of the present invention. As shown in FIG. 5, the array substrate 2b in this variation embodiment differs from the array substrate 2a illustrated in FIG. 3 in that the traces 18 in the variation embodiment may have the same length in the active region AA. For example, the trace 18a and the trace 18b have the same length along the second direction D2. Under this circumstance, the trace 18a in the active region AA may overlap the bridging sections 162 of other scan lines, which are not electrically connected thereto, in the top view of the array substrate 2b. Therefore, the coupling capacitance of each of the scan lines 16 with other scan lines 16 may be the same, and the coupling capacitances of the scan lines can be unified, so as to minimize abnormality of the display device and improve displaying quality. Other portions of the array substrate 2b shown in FIG. 5 may be similar to or the same as those included in the array substrate described above or below, and thus it will not be described redundantly.

[0046] FIG. 6 schematically illustrates a top plan view of an array substrate according to another variation of the second embodiment of the present invention. As shown in FIG. 6, the array substrate 2c in this variation embodiment differs from the array substrate 2a illustrated in FIG. 3 in that the data lines 20 in the variation embodiment may be disposed at respective left sides of corresponding columns of pixel electrodes 14, i.e., the columns of pixel electrodes 14 electrically connected thereto. In some embodiments, the traces 18 may be getting longer along the first direction D1. For example, longer traces 18 are closer to the second side S3 of the active region AA, but it is not limited thereto. Other portions of the array substrate 2c shown in FIG. 6 may be similar to or the same as those included in the array substrate described above or below, and thus it will not be described redundantly.

[0047] FIG. 7 schematically illustrates a top plan view of an array substrate according to a third embodiment of the present invention. FIG. 8 schematically illustrates a cross-sectional view of FIG. 7 taken along a line C-C'. As shown in FIG. 7, the array substrate **3a** in this embodiment differs from the array substrate **2a** illustrated in FIG. 3 in that at least one of the traces **18** in this embodiment may overlap at least one of the pixel electrodes **14** in the top view of the array substrate **3a**. For example, the trace **18a** is connected to the scan line **16a**, and the pixel electrode **14b** is disposed between the scan line **16a** and the first side **S1** of the active region **AA**, so the trace **18a** crosses the pixel electrode **14b**, and so forth. Under this circumstance, the at least one of the scan lines **16** may further include at least one bridging section **163** in addition to the line sections **161** and the bridging sections **162**, and in the top view of the array substrate **3a**, the bridging section **163** may be overlapped with and insulated from one of the traces **18**. For example, the scan line **16b** may further include one bridging section **163**, which overlaps the trace **18a** electrically connected to the scan line **16a**. The scan line **16c** may include two bridging sections **163**, which overlap the trace **18a** and the trace **18b**, respectively, and so forth.

[0048] In the embodiment as illustrated in FIG. 7, one of the common lines **24** is disposed between one of the traces **18** and one of the data lines **20** in the top view of the array substrate **3a**, thereby alleviating the influence of the signal transmitted through the trace **18** on the data signal transmitted through the data line **20**, but it is not limited thereto. In some embodiments, the positions of the bridging sections **163** may vary with the positions of the traces **18**, and it is not limited to the configuration shown in FIG. 7.

[0049] As shown in FIG. 8, the bridging sections **162** and the bridging sections **163** are formed of the same conductive layer, e.g., the conductive layer **C3**, but it is not limited thereto. In some embodiments, other portions of the array substrate **3a** shown in FIG. 7 and FIG. 8 may be similar to or the same as those included in the array substrate described above or below, and thus it will not be described redundantly.

[0050] FIG. 9 schematically illustrates a top plan view of an array substrate according to a variation of the third embodiment of the present invention. As shown in FIG. 9, the array substrate **3b** in this variation embodiment differs from the array substrate **3a** illustrated in FIG. 7 in that the bridging sections **163** and the bridging sections **162** are formed of different conductive layers. For example, the bridging sections **163** and the data lines **20** are formed of the same conductive layer **C2**. Since the conductive layer **C2** may include metal, the influence of the bridging sections **163** on the resistances of the scan lines **16** can be alleviated, so as to unify the resistances of the scan lines. In some embodiments, other portions of the array substrate **3b** shown in FIG. 9 may be similar to or the same as those included in the array substrate described above or below, and thus it will not be described redundantly.

[0051] FIG. 10 schematically illustrates a top plan view of an array substrate according to another variation of the third embodiment of the present invention. As shown in FIG. 10, the array substrate **3c** in this variation embodiment differs from the array substrate **3a** illustrated in FIG. 7 in that each of the scan lines **16** may include a plurality of bridging sections **163**, and the scan lines **16** may have the same number of bridging sections **163**, so as to unify the resistances of the scan lines. In an embodiment, the bridging sections **163** of the scan lines **16** may have the same length in the first direction **D1**, but it is not limited thereto. Under this circumstance, the traces **18** in the active region **AA** may optionally be the same, so that one trace **18** may overlap one of the bridging sections **163** of each scan line **16**, which is not electrically connected thereto, in the top view of the array substrate **3c**, thereby unifying the coupling capacitance of the scan lines **16**. For example, one of the bridging sections **163** of the scan line **16a** may overlap the trace **18b** electrically connected to the scan line **16b**. In some embodiments, other portions of the array substrate **3c** shown in FIG. 10 may be similar to or the same as those included in the array substrate described above or below, and thus it will not be described redundantly.

[0052] To sum up, in the array substrate according to the present invention, since the traces

electrically connected to the scan lines extend from the first side of the active region to the peripheral region, the number of traces located in the peripheral region at the second side and the third side of the active region can be greatly reduced, so that the display device can have a narrow border. Moreover, the traces and the data lines may be formed of different conductive layers to reduce the coupling capacitances between the traces and the data lines, so as to minimize abnormality of the display device. Furthermore, by designing the scan lines with the same number of bridging sections, the resistances of the scan lines may be unified, so as to improve the displaying quality of images.

[0053] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. An array substrate, comprising: a substrate; a first pixel electrode and a second pixel electrode disposed on the substrate and arranged along a first direction; a first scan line and a second scan line disposed on the substrate and extending along the first direction, wherein the first pixel electrode and the second pixel electrode are disposed between the first scan line and the second scan line; a first trace and a second trace disposed on the substrate and extending along a second direction different from the first direction, wherein the first trace is electrically connected to the first scan line, the second trace is electrically connected to the second scan line, the first trace crosses and is electrically insulated from the second scan line, and the first trace and the second trace are formed of a first conductive layer; and a data line disposed on the substrate and extending along the second direction, wherein in a top view of the array substrate, the data line is disposed between the first pixel electrode and the second pixel electrode, the data line crosses the second scan line, the data line is formed of a second conductive layer different from the first conductive layer, and the data line is disposed between the first trace and the second trace.
2. The array substrate according to claim 1, wherein the second scan line comprises at least two line sections and at least one bridging section, the at least one bridging section is electrically connected between the at least two line sections, the at least one bridging section is formed of a third conductive layer different from the first conductive layer and the second conductive layer, the at least one bridging section overlaps the first trace in the top view of the array substrate, and the at least two line sections are formed of the first conductive layer.
3. The array substrate according to claim 1, further comprising a first switch element and a second switch element disposed on the substrate, wherein a gate electrode, a source electrode and a drain electrode of the first switch element are electrically connected to the first scan line, the data line and the first pixel electrode, respectively, a gate electrode, a source electrode and a drain electrode of the second switch element are electrically connected to the second scan line, the data line and the second pixel electrode, respectively, and the source electrode of the first switch element overlaps the first trace in the top view of the array substrate.
4. The array substrate according to claim 1, further comprising: a third pixel electrode and a fourth pixel electrode disposed on the substrate and arranged along the first direction, wherein the first pixel electrode and the third pixel electrode are arranged along the second direction; a third scan line and a fourth scan line disposed on the substrate and extending along the first direction, wherein the third pixel electrode and the fourth pixel electrode are disposed between the third scan line and the fourth scan line; and a third switch element and a fourth switch element disposed on the substrate, wherein a gate electrode, a source electrode and a drain electrode of the third switch element are electrically connected to the third scan line, the data line and the third pixel electrode, respectively, and a gate electrode, a source electrode and a drain electrode of the fourth switch

element are electrically connected to the fourth scan line, the data line and the fourth pixel electrode, respectively, and wherein in the top view of the array substrate, the source electrode of the third switch element overlaps the first trace, and the source electrode of the fourth switch element overlaps the second trace.

5. The array substrate according to claim 4, wherein the third scan line comprises at least two line sections and at least one bridging section, the at least one bridging section is electrically connected between the at least two line sections, and the at least one bridging section overlaps and is electrically insulated from the first trace and the second trace in the top view of the array substrate.

6. An array substrate, comprising: a substrate; a first scan line and a second scan line disposed on the substrate and extending along a first direction, wherein each of the first scan line and the second scan line comprises a plurality of line sections and a plurality of first bridging sections; a first trace and a second trace disposed on the substrate and extending along a second direction different from the first direction, wherein the first trace is electrically connected to the first scan line, and the second trace is electrically connected to the second scan line; and a data line disposed on the substrate and extending along the second direction, wherein the data line crosses one of the first bridging sections of the first scan line and one of the first bridging sections of the second scan line, wherein a number of the first bridging sections of the first scan line is the same as a number of the first bridging sections of the second scan line, and wherein in a top view of the array substrate, the one of the first bridging sections of the second scan line in the data line overlaps the data line and the first trace.

7. The array substrate according to claim 6, wherein the line sections are formed of a first conductive layer, the data line is formed of a second conductive layer, the first bridging sections are formed of a third conductive layer, and the first trace and the second trace have a same length in the second direction.

8. The array substrate according to claim 6, further comprising: a first pixel electrode and a second pixel electrode disposed on the substrate and arranged along the second direction; and a first switch element and a second switch element disposed on the substrate, wherein the first pixel electrode is electrically connected to the data line through the first switch element, and the second pixel electrode is electrically connected to the data line through the second switch element.

9. The array substrate according to claim 6, wherein the first trace overlaps the second pixel electrode in the top view of the array substrate, the second scan line further comprises at least one second bridging section, and the first trace crosses the at least one second bridging section of the second scan line in the top view of the array substrate, and wherein the first bridging sections and the at least one second bridging section of the second scan line are formed of a same conductive layer or formed of different conductive layers.

10. The array substrate according to claim 9, wherein the first scan line further comprises at least one second bridging section, a number of the at least one second bridging section of the first scan line is the same as a number of the at least one second bridging section of the second scan line, and the at least one second bridging section of the first scan line overlaps the second trace.
